MantraCore Solutions – MCS

" Smart Solutions for Greener, Connected Communities."

VHEcoPoint: A Smart Waste Segregation Station

with QR-Based Incentive System for Victorian Heights Subdivision

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2026

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Chapter 1. Startup Branding

1.1 Company Name

The official company's name is MantraCore Solutions (MCS). The name was inspired by our capstone team name, "Team Mantra," allowing us to preserve the identity and values the company established throughout the development of the group's inception. We combined "Mantra" with "Core" to represent technology as the foundation of every solution created, while "Solutions" reflects the commitment to solving real-world problems through innovative and practical Internet-of-Things (IoT) and web-based technologies.

The name directly relates to the company's mantra, which aims to empower residential communities with smart services. Therefore, VHEcoPoint: A Smart Waste Segregation Station with QR-Based Incentive System was established, together with the VictorianPass web platform. These technologies work together to promote sustainable waste management and improve community engagement. MantraCore Solutions is short, professional, and easy to remember, making it appropriate for the target market, including homeowners' associations (HOAs), local government units (LGUs), and residential communities. Its broad and professional identity also allows the startup to expand into future smart community solutions beyond its initial product.

Beyond its practical branding value, the name MantraCore Solutions carries a deeper significance for the founding team. It represents the transition of "Team Mantra" from a student-led capstone group into a startup with a clear market identity and long-term direction. This continuity reinforces the company's commitment to the principles it was built on, including innovation, sustainability, and community-centered service, while signaling to stakeholders that MCS is not merely a school project, but a scalable venture capable of delivering real, lasting solutions to the communities it serves.

1.2 Company Logo

Figure 1: Company Logo

Meaning of the Logo

The logo centers around a bold, gold "M" with a dotted line forming a peaked roof above it, evoking the shape of a house a direct nod to the residential communities MantraCore Solutions we aim to serve. While dotted lines can emphasis of flow which guides the users of what they can receive when finding our assistance, A sweeping white cursive "C" overlays the "M," together forming a compact "MC" monogram (MantraCore) that carries the brand name at a glance. The house-like silhouette reinforces the company's community and home-focused identity.

Color Psychology

- Deep forest green (background): Grounds the mark, feels premium and stable, and nods to the sustainability angle behind the company's work without being an overly literal "leaf green." Green is also the universally recognized color of nature and environmental health.
- Gold/Yellow ("M," dotted roofline, "MANTRA" text): Reinforces the house/roof motif and the bold "M," signaling prestige, value, and excellence and gold is also the color most associated with rewards and points, tying naturally to an incentive-driven product line.
- White (cursive "C" swoosh, "CORE" text, "SOLUTIONS" bar): Provides contrast and legibility against the dark background, giving the mark a clean, confident focal point where the "C" cuts across the "M," and grounding "SOLUTIONS" as the company's clear, deliverable-focused tagline element.

Font

The wordmark uses The Seasons, a bold, condensed small-caps serif typeface, for both "MANTRACORE" and "SOLUTIONS" the main wordmark set in full weight for strength and presence, while "SOLUTIONS" underneath is set in a lighter, wide-tracked variation for contrast. The Seasons serif gives the company name a strong, professional, and timeless look, while the lighter tracking on "SOLUTIONS" adds a modern, approachable feel. Together, they reflect a startup that is both reliable and community focused.

Symbolism

- The gold "M" topped with a dotted roofline forms the silhouette of a house, symbolizing home, community, and the residential subdivisions MantraCore Solutions is built to serve.
- The cursive "C" swoosh cutting through the "M" represents "Core," its fluid, dynamic stroke contrasting against the rigid geometric lines of the house-and-M shape, symbolizing agility and innovation working within a foundation of structure and reliability.

- Together, the bold "M" and cursive "C" form a unified "MC" monogram, the clearest possible visual shorthand for MantraCore.
- The white "SOLUTIONS" bar beneath the wordmark acts as a visual anchor, grounding the company's actual deliverable (solutions) beneath the more symbolic house-and-monogram mark above it.

1.3 Tagline

"Smart Solutions for Greener, Connected Communities."

This tagline was chosen because it captures the two pillars of business in a single line: 'Smart Solutions' speaks to the Internet-of-Things (IoT) and web-based technology, while 'Greener, Connected Communities' speaks to the environmental and social impact that aim to create. It directly reflects the mission of serving residential communities through practical, sustainable technology, and reinforces brand identity as a company that blends innovation with genuine community benefit rather than pursuing technology for its own sake.

1.4 Vision

To become a leading provider of innovative IoT and smart community technologies that empower communities to become greener, smarter, and more connected through sustainable digital solutions.

1.5 Mission

MantraCore Solutions develops sustainable IoT and web-based technologies that solve real community challenges, delivering smarter services and greater value for homeowners' associations, subdivision administrators, and residential communities.

1.6 Core Values

- **Innovation** – To continuously develop creative IoT and digital technologies that solve real community challenges and improve everyday living. At MantraCore Solutions, innovation means constantly refining our systems, from VictorianPass's entry security features to VHEcoPoint a waste segregation and incentive system. So that our solutions stay relevant, efficient, and responsive to the evolving needs of the communities we serve.
- **Sustainability** – We design environmentally responsible solutions that promote proper waste management and long-term community development. This value drives every design decision we make, ensuring that our technologies not only solve immediate waste management challenges but also contribute to lasting environmental impact and support communities in building greener, more sustainable habits.
- **Integrity** – We build trust by delivering reliable, transparent, and ethical technology solutions. We hold ourselves accountable to honest communication with our clients and stakeholders, from accurate point-tracking systems to clear reporting, so that every partner and resident can rely on MCS with confidence.

- **Community-Centered Service** – We prioritize the needs of communities by creating practical technologies that improve quality of life. Every feature we build, from VictorianPass's entry security features to VHEcoPoint a waste segregation and incentive system, is shaped around the real experiences and feedback of the residents and administrators who use it daily.
- **Excellence** – Committed to maintaining high standards in every product, service, and innovation we deliver. We continuously test, refine, and improve our systems before and after deployment, ensuring that MantraCore Solutions consistently delivers dependable, high-quality technology that communities can trust for the long term.

Chapter 2. Project Overview

2.1 Introduction

MantraCore Solutions (MCS) is a technology startup focused on developing innovative, practical, and community-centered solutions that address real-world problems by combining modern technologies with efficient systems to improve community services, promote sustainability, and encourage responsible participation among residents. One of the company's proposed solutions is VHEcoPoint, a Smart Waste Segregation Station with QR-Based Incentive System designed specifically for residential communities such as Victorian Heights Subdivision. The project addresses common waste management challenges, including improper segregation, inefficient disposal practices, and limited resident participation in recycling activities, by introducing a technology-assisted approach that encourages residents to properly dispose of recyclable materials while receiving incentives. VHEcoPoint integrates a physical waste segregation station with the VictorianPass web platform and uses a Raspberry Pi 4B as its primary processing unit, an ESP32-CAM for image-based material recognition, a GM65 QR Code Scanner for resident identification, an inductive sensor for material detection, and an HX711 load cell for weight measurement. The process begins when a resident scans their assigned QR code using the GM65 QR Code Scanner, allowing the system to identify the resident before recyclable materials are deposited into the station. The system then assists in recognizing the material through the inductive sensor and ESP32-CAM while the HX711 load cell measures its weight. The collected data is processed and recorded through the VictorianPass platform, where applicable reward points are computed and credited to the resident for accumulation and use toward available community incentives. The integration of VHEcoPoint with VictorianPass also provides subdivision administrators with centralized digital records of waste disposal activities, collected materials, weights, resident participation, and earned points, reducing reliance on manual records and supporting data-driven monitoring and evaluation. More importantly, VHEcoPoint aims to encourage residents to develop better waste segregation habits by making recycling more convenient, organized, and engaging through automated identification, material recognition, weight measurement, digital recording, and an incentive mechanism. This chapter presents the overall project overview of VHEcoPoint, including its background, purpose, target users, proposed features, operational process, technical components, and innovative aspects, while explaining how its hardware and software components work together to support a more organized, technology-driven, and sustainable approach to recyclable waste management in Victorian Heights Subdivision.

2.2 Problem Statement

Victorian Heights Subdivision faces several challenges in waste management that affect environmental sustainability, operational efficiency, and resident participation. Based on observations and discussions with subdivision representatives, issues such as improper waste segregation, limited monitoring, and low resident participation were identified. These problems are mainly caused by the lack of an organized waste management system, limited monitoring of waste activities, and insufficient motivation for residents to practice proper waste disposal.

Without an effective solution, these challenges may continue to affect the cleanliness of the community, the efficiency of waste collection, and resident participation in recycling activities. Therefore, there is a need for a technology-based and community-focused waste management solution that can encourage proper segregation, monitor recycling activities, and motivate residents through incentives. VHEcoPoint is proposed to help address these concerns by providing a more organized, monitored, and engaging approach to waste management

Improper Waste Disposal Practices. Many residents dispose of recyclable materials together with general waste, resulting in mixed waste that significantly reduces recycling opportunities and increases the volume of waste sent to landfills. This practice is largely influenced by convenience and the absence of an efficient and accessible waste segregation system within the subdivision. Without clearly labeled compartments or automated sorting mechanisms, residents have little guidance on how to properly separate their waste. Consequently, valuable materials that could have been repurposed or recycled end up in landfills, contributing to environmental pollution and accelerating the depletion of landfill space. This improper disposal habit not only wastes resources but also places an unnecessary burden on the entire waste management chain, from collection to final disposal.

Accumulation of Recyclable Materials. Residents often store recyclable materials such as PET bottles, aluminum cans, paper, and cardboard inside their homes due to the lack of a convenient and reliable collection system. Over time, this practice results in significant household clutter, as recyclables accumulate in storage areas, kitchens, or garages. The prolonged storage of these materials attracts pests like cockroaches and rodents, creates potential fire hazards—especially when paper and cardboard are involved—and reduces the available living space within households. Furthermore, the deterioration of stored materials can render them unrecyclable, defeating the purpose of saving them in the first place. This issue highlights the urgent need for a system that provides residents with a regular, hassle-free way to dispose of their recyclables without compromising their home environment.

Increased Workload for Subdivision Personnel. Without a systematic waste segregation process in place, subdivision maintenance personnel are required to manually sort recyclable materials after they are collected from amenity areas and communal spaces. This manual sorting process is extremely time-consuming, inefficient, and diverts personnel away from other essential maintenance duties. Moreover, it exposes workers to significant health and safety risks, including contact with sharp objects, biohazards, and toxic substances that may have been improperly disposed of alongside recyclable materials. The physical strain and occupational hazards associated with manual sorting not only affect worker morale but also increase the likelihood of workplace injuries and associated costs. This unsustainable practice underscores the need for an automated solution that minimizes human intervention in the segregation process.

Lack of Monitoring and Tracking. The subdivision currently operates without a centralized system for monitoring waste collection activities, tracking resident participation, or recording the quantity and types of recyclable materials collected. As a result, administrators have limited access to reliable data needed to evaluate the effectiveness of existing waste management initiatives. This data deficiency makes it difficult to identify trends, measure improvements, or allocate resources efficiently. Without accurate records, the subdivision cannot determine which households are actively participating in segregation efforts or assess the overall environmental impact of their programs. The absence of a tracking mechanism essentially blinds administrators to the true state of waste management within the community, preventing informed decision-making and strategic planning.

Low Resident Participation. The absence of incentives or recognition discourages residents from actively participating in proper waste segregation activities. For many households, waste disposal is viewed as a mundane obligation rather than a meaningful community activity that benefits the environment and the neighborhood as a whole. Without a reward mechanism that acknowledges and values their efforts, residents lack the motivation to change their disposal habits or invest time in sorting their waste. This disengagement perpetuates the status quo, where improper disposal remains the default behavior. The lack of a gamified or incentive-driven system means that even environmentally conscious residents may eventually lose interest if their contributions go unnoticed and unrewarded.

These interconnected challenges create an inefficient and unsustainable waste management process that negatively affects residents, maintenance personnel, and subdivision administrators alike. Addressing these issues requires a solution that goes beyond traditional awareness campaigns and introduces a holistic approach combining automation, real-time monitoring, data-driven decision-making, and incentive-based participation. Only through such an integrated system can Victorian Heights Subdivision break the cycle of inefficiency and build a cleaner, safer, and more environmentally responsible community for all its residents.

2.3 Proposed Solution

MantraCore Solutions proposes VHEcoPoint, a Smart Waste Segregation Station with QR-Based Incentive System that serves as an integrated hardware and software solution for automating waste segregation, monitoring resident participation, and promoting environmentally responsible behavior through a reward-based incentive program.

The proposed solution consists of two major components:

Hardware Component: VHEcoPoint Smart Waste Segregation Station

- **IoT-Enabled System:** VHEcoPoint is powered by a raspberry Pi 4b microcontroller, which serves as the system's primary processing unit for both hardware and software that enables real-time communication with the VictorianPass web platform.
- **QR Code Identification:** A GM65 QR code scanner identifies each resident before recyclable materials are deposited. Every resident is assigned a unique QR code linked to their VictorianPass account, ensuring that every transaction is secure and accurately recorded.
- **Automated Material Classification:** VHEcoPoint uses an inductive sensor to detect metallic materials, while the ESP32-CAM camera classifies recyclable materials such as paper and plastic with the use of machine learning with YOLOv11. This automated process minimizes manual waste sorting and improves operational efficiency.
- **Weight-Based Measurement Module:** An HX117 loadcell module measures the weight of deposited recyclable materials. The materials weigh will convert to points which the users can accumulate for amenity ussage discount.

Operational Process

1. Resident Identification – The resident scans their personal QR code using the GM65 QR code scanner.
2. Waste Deposit – The resident deposits recyclable materials, such as paper, PET plastic, bottles or aluminum cans into the VHEcoPoint Smart Waste Segregation Station.
3. Automated Material Classification – The station automatically identifies the type of recyclable material using the inductive sensor and the ESP32-CAM camera while streaming YOLOv11.
4. Weight Measurement – The HX117 loadcell sensor measures the weight of the deposited recyclable material.
5. Reward Point Computation – The system automatically computes reward points based on the material type and weight, then credits the points to the resident's VictorianPass account.
6. Data Recording – Transaction details are transmitted to the VictorianPass web platform for real-time monitoring, reporting, and record management.

This integrated solution addresses the identified waste management challenges by providing convenience through automation, accountability through QR-based monitoring, and motivation through a reward-based incentive program. In addition, the system generates valuable data that enables subdivision administrators to evaluate program effectiveness, improve waste management strategies, and encourage greater community participation.

2.4 Target Users

The proposed solution is designed to benefit multiple stakeholders within the residential community.

Primary Users: Residents of Victorian Heights Subdivision

Residents are the primary users of VHEcoPoint. The system provides a convenient and automated waste disposal process while rewarding environmentally responsible behavior through a points-based incentive program. The QR-based identification system ensures that every transaction is accurately recorded, allowing residents to monitor their accumulated reward points and track their contributions to community sustainability efforts

Secondary Users: Homeowner's Association (HOA)

Subdivision administrators and HOA officers oversee the operation of the system through the VictorianPass web platform. They can monitor resident participation, review waste collection statistics, generate reports, and evaluate the effectiveness of waste management initiatives. The collected data supports informed decision-making, policy development, and continuous improvement of community waste management programs.

Supporting Stakeholders: Subdivision Maintenance Personnel

Subdivision maintenance personnel benefit from reduced manual waste segregation, allowing them to perform waste collection more efficiently and safely. The automated segregation process minimizes the handling of mixed waste, reduces workload, and promotes a cleaner and more organized waste collection system.

Overall, the proposed solution creates value for residents, subdivision administrators, and maintenance personnel by improving waste management efficiency, encouraging active community participation, and promoting environmental sustainability through smart technology and incentive-based engagement.

Chapter 3. Innovation

3.1 Innovation Statement

The MantraCore Solutions as a start-up strives to explore, design and innovate new ways to address community problems in regards of waste management with the use of technology. In this section, the following questions will be answered about the start-up:

- What is the name of the startup?

  The name MantraCore is derived from the developers' capstone team identity, which is "Team Mantra." The developers decided to add the word "Core" to emphasize that as a startup center around technology that focuses on solving community-wide problems, with the use of IoT-based innovations. As for "Solutions", it signifies that MantraCore exists to solve future clients' real and practical solutions to their problems. The developers believe that with the use of technology there are always solutions for greener and connected communities.
- What problem does it address?

  MantraCore Solutions Addresses real community challenges in regards of proper waste management and residential engagement. The startup believes in the exploration, designning and innovative approaches with the use of technology to improve quality of life and community welfare.

- What innovative solution does it provide?

  One of the Innovative is the VHEcoPoint a smart waste segregation station with QR-based Incentives system that integrates Quick-Response Code for user identification, material segregation feature and weight-based rewarding system to improve waste segregation and community engagement through internet of things technologies.
- Who are the target users or customers?

  The target users and customers of MantraCore Solutions are the homeowners' associations (HOA), residents, and local communities.

3.2 Design Thinking Summary

In this section shows the summarization of the design thinking process that are applied into making MantraCore Solutions' innovative systems. This approach is important for MantraCore as it can be used for making sure that the development team will not deviate with the planning. This will focus on user-centered approach in solving problem which focuses on understanding the needs of the client. This section will include understanding pain-points of the target users, identifying the problem, generate solution to solve the problem. With the ongoing increase of waste segregation malpractices and abundance of general waste buildup, the company came up solution using Internet-of-Things (IoT) to make ease of these problems such as; VHEcoPoint and VictorianPass web system. Both systems complement each other to solve resident's problem of security and waste management.

Empathize

The developers conducted interviews and consultations with subdivision administrators and residents on Victorian Heights Subdivision to understand existing waste management malpractices and the challenges experienced within their community. The gathered information revealed that homeowners often purchase several water bottles packages to be supplied on a weekly basis resulting in abundance of empty bottles, accumulation of stored recyclables, increase in waste segration malpractices and increased workload for subdivision personnel in regards of amenity waste collection.

Findings:

Operations Department

Commented [TV1]: estimate gano kareaming bottles each household

- Recyclable waste hauling volume increases as residents tend to buy large quantities of water bottles for daily use in a week (Example: 2 boxes of 'SUMMIT' drinking water (12 pieces of 1000ml) every other day.)
- Difficulty in proper waste segregation, as recyclable materials were mixed with another non-recyclable (biodegradable and nonbiodegradable), that is detrimental to environment – causes pollution as plastic tend to not dissolve decades or centuries depends on the material.

Finance Department

- Disposal expenses are Increased, while mixed with other waste its losses its value.
- Cluttered materials increase amenities means an increase in cleaning materials expenses.
- The department does not capitalize on the bottle build up, as a lot of the plastic bottles were thrown mixed with the other materials in waste collection day, which provides no passive income for maintenance.

Amenities & Events Department

- Customers often leave clutter material after usage of an amenity, which adds work to the staff.

Community Relations / Administration

- Lack of community awareness about sustainability and health well-being.
- Lack of engagement on proper segregation and waste collection

Define

According to the interview conducted by the MantraCore Solutions, Victorian Heights Subdivision is experiencing ongoing waste management issues caused by improper waste disposal, recyclable waste stockpiling, lack of awareness in regards of waste segregation and the absence of a monitoring system for waste collection activities. These challenges reduce waste collection efficiency, contribute to household and community clutter, and increase the workload of subdivision management personnel – to add, recyclable wastes such as pet bottles are mixed with other waste contributes to disrupting decomposition processes (IERE,2025). The lack of proper waste management can be detrimental not only to health but also sustainability as these materials can offer value to subdivisions' maintenance.

Problem Statement

- Improper waste disposal practices among residents
- Prolonged stockpiling of recyclable materials such as PET bottles, aluminum cans, paper, and small to medium small to medium Cardboard boxes.
- Accumulation of household clutter ("abubot") including old documents and unused materials.
- Additional workload for subdivision management in handling scattered waste.
- Absence of monitoring and tracking system for waste collection and disposal.
- Lack of data to evaluate waste management effectiveness and resident compliance.

Ideate:

Based on the empathize stage of the study, Victorian Heights Subdivision is experiencing ongoing waste management issues caused by improper waste disposal, recyclable waste stockpiling, lack of awareness in regards of waste segregation and the absence of a monitoring system for waste collection activities. These challenges reduce waste collection efficiency, contribute to household and community clutter. After identifying the primary problems, the developers brainstormed several potential technology-driven solutions that could improve waste management while encouraging greater community participation.

Proposed Ideas:

- Develop an IoT-based Smart Waste Segregation Station capable of identifying recyclable waste categories Such as (Paper, Plastic and Aluminum Cans).
- Integrate QR Code identification to uniquely identify residents before waste disposal.
- Implement weight-based reward computation to reward according to material and usage.
- Develop a web-based management system for monitoring resident participation, waste collection records, and generated reports.
- Provide real-time recording of waste disposal transactions and reward points.
- Utilize sensor (Inductive sensor) and object recognition technologies with ESP32-CAM (YOLOv11) to automate recyclable waste classification.
- Generate reports to assist subdivision administrators in evaluating waste management performance (Excel file).

Prototype

The developers designed and developed an initial working prototype consisting of both hardware and software components that demonstrate the core functionalities of the proposed system.

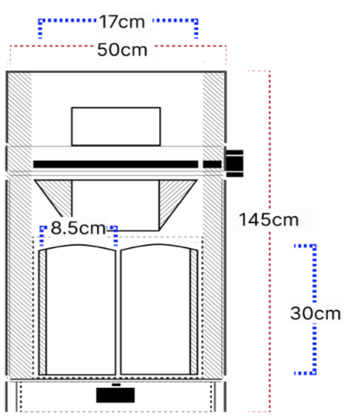

Figure 2: VHEcoPoint System Concept

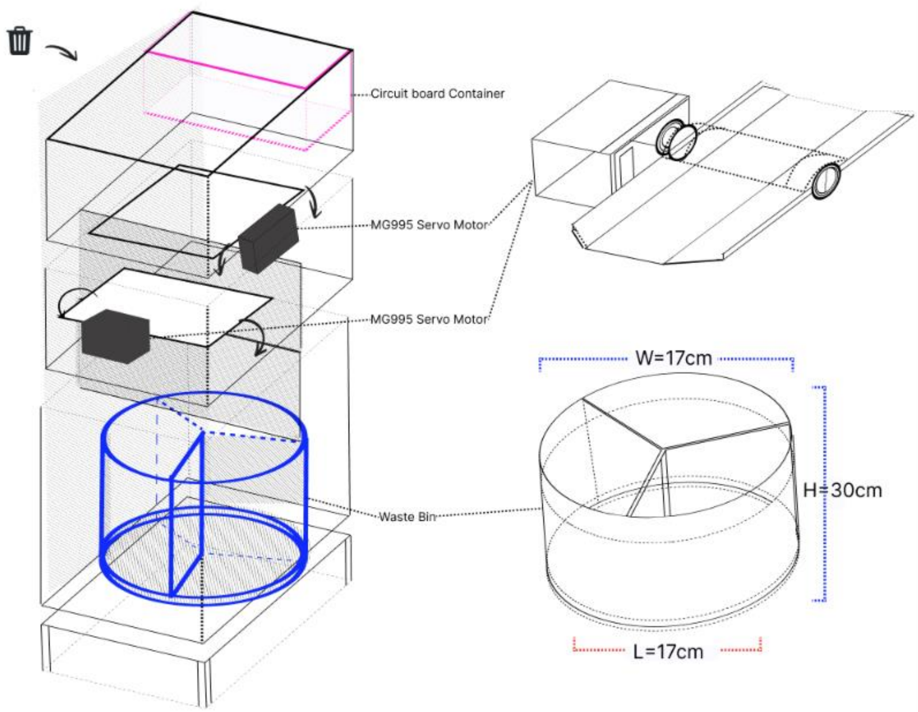

Figure 3: VHEcoPoint Prototype

Prototype Description:

The VHEcoPoint: Smart Waste Segregation Station with QR-Based Incentive System for Victorian Heights Subdivision is designed by MantraCore Solutions to support Sustainable Development Goal SDG 11: Sustainable Cities and Communities and SDG 12: Responsible Consumption and Production, which aims to minimize adverse impact on human health and the environment. This system utilizes IoT-based hardware components such as microcontroller, sensors and servo motors to deliver reliable and efficient waste segregation system. In a related capstone study of (Carolino, E. B., et al., 2024), refers a system that utilizing sensors for waste segregation and monitoring. a capable system that segregates materials based on its color and weight classes – promptly closing the trash bin after the item was thrown to the trash bin after detection. Another Capstone Project named WasteWise developed by (De Castro, et al., 2026) etc. Designed as an automated waste disposal that addresses challenges of improper waste disposal and low recycle motivation among school environments. With this, MantraCore Solutions have a baseline on how the system could be built.

Technical architecture has follows layered design of hardware, presentation, backend layer:

In the hardware layer, Raspberry Pi 4b was utilized as the main microcontroller to manage and send signal to the embedded system. The microcontroller was powered by being connected to a Type C charger that can supply 3A (3 amps) output min, any lesser than 3A is below the required power capacity of the microcontroller. The Raspberry Pi 4b manages the sensors which are significant in sorting waste process; Inductive Sensor is used to detect metal (aluminum cans), ESP32-CAM for non-metal (plastic and paper). These materials are difficult to differentiate as they have same qualities depending on composition and material to which they are made. The developers decided to use ESP32-CAM as the streaming medium, to implement YOLOv11 machine learning to the system making ease of the material detection problem. What is YOLOv11? According to (Gallagher, J., 2024), it's a computer vision model that allows object detection, segregation and classification. The development team aims to achieve at least 0.90 in mean average precision (mAP), which is a metric to evaluate detection performance. While on image gathering at least 2000 images. This will ensure that the system can differentiate both classes without giving wrong reading. Furthermore, the system will be implemented with Two (2) MG995r as a means as a coverer for the materials after detection, which should redirect them to their respective waste bins.

At the presentation layer, The VictorianPass Website is designed for Victorian Height Subdivision. This is the core system that manages user Interaction within a web setting. This is where the admin and the residents will monitor their activity and points accumulated remotely. At the resident side, they will download a QR code provided to them by the website to access the IoT system. before using the waste segregation station, the user is required to scan the QR code into a GM65 QR / Barcode scanner that will verify their access. With the disposal of waste, their points will be based on what waste thrown out accumulating points to be presented on an LCD 1602 Display and web user dashboard. For admin, they can monitor user activity and how much waste the system has accumulated on the system. Ensuring cleanliness and engagement through proper waste management.

For the backend, this is the backbone of the system - that handles the logic of how the system will gather the data from the sensors and then connect it to the website VictorianPass. which the developers of MantraCore Developed which is made with PHP, HTML and deployed on Hostinger web hosting services. This will include data validation, user tracking, and point accumulation storage. This acts as the core for data synchronization and storage between the physical system and the website.

In Raspberry Pi 4b, Linux was written the Primary operating system (OS) to control the whole operation on the microcontroller, while instead of C++ as the language for hardware configuration it was coded on Python language. In conclusion, this section can help the developers map out how does the system can be executed into to be plausible.

Test

The prototype will be presented to the stakeholders after the completion of system development, to evaluate its functionality, usability, and practicality. With giving them access to a program that demonstrate on how system works and proper waste management. Feedback gathered during testing will be crucial to the developers as this can be used for the improvement of the system and identify areas for risk before further development.

3.3 Proposed Innovative Features

Table 1: Proposed Innovative Features

Table 2: Proposed Innovative Features

Feature	Description	Benefits to the user
Feature 1	With the GM65 QR code scanner, the system can scan QR code to Identify the end user	Added system security and web account connection for incentives.
Feature 2	Reward points given depend on the material sensed by the HX711 loadcell module that will be counted on the user's account on "VictorianPass".	Encourages residents to participate in proper waste segregation while receiving incentives
Feature 3	Segregates material based on category (paper, plastic, and metal) by means of integrating Inductive sensor (aluminum cans) and ESP32-CAM with stream interference of YOLOv11 (plastic and paper)	User will benefit on automated waste segregation, removing the need for manual sorting.
Feature 4	The system records each session to a website called "VictorianPass."	The user can have an overview of their activities and incentives on the website. The incentives can be used as a discount for amenity usage.

3.4 Innovation Value

The Value of MantraCore Solutions comes in the form of innovations integrated with IoT-based technologies which turn challenges into opportunities. With the goal of creating systems that are environmentally friendly while further elevating user's quality-of-life.

Customer Value

MantraCore solutions always seek to value customer's satisfaction by means of making clients worth their time, effort, investment. Ensuring users can trust the product and perception of the company itself. With innovations such as VHEcoPoint: smart waste segregation station with QR-based incentives system and VictorianPass, which provides customer value as an alternative for traditional waste segregation methods. Utilizing IoT technology, without manual segregation of waste, the system's task is to segregate them to its waste class. The user then receives incentives based on activity through a website "VictorianPass" to which in return can be accumulated for amenity usage discount. This system is designed to encourage customers to be aware of proper waste disposal, and cleanup engagement.

Social Value

For social value, MantraCore Solutions ensures that every service provided can ensue positive impact not only for the environment but also our client's well-being. That aims to encourage and promote awareness of improper waste malpractices such as; prevention of amenity littering and inefficient recycling practices that contributes to the accumulation of waste. with Fostering sustainable waste management practices, MantraCore Solutions can develop innovation contributes to a cleaner, and healthier client and subdivision's welfare. In the long run, setting sights on encouraging users to develop responsible waste disposal habits and recognize their role in protecting the environment. These efforts support the United Nations Sustainable Development Goals (SDGs), particularly SDG 11: Sustainable Cities and Communities and SDG 12: Responsible Consumption and Production

Economic Value

MantraCore Solutions aims to provide economical value through the system that they provide by allowing efficient solution for the clients to use. With VHEcoPoint, users can have the opportunity to help the environment, while regrating waste such as; paper, Pet Bottle and Aluminum cans that may lost its value if mixed with biodegradable and a massive detriment to the biodegradable breakdown process. The income generated from these recyclables can help provide financial support for the maintenance of community amenities and waste management activities. In this way, the system also has weight-based incentive rewarding system to promote using the system, which increases the load of recyclable materials that can be sold off for economic value. The system helps minimize waste while transforming waste disposal and sorting concern into a potential economic resource. Which creates a cycle of users ere encourage to recycle, waste materials retain its value and can contribute financial support in maintaining community welfare.

Competitive Value

Similar innovations that had been made over the years, similar goal to promote healthier and engaging waste management systems. Which aims to sort biodegradable and non-biodegradable materials preventing abundance of mixed general waste that causes massive landfills. An example of this are the Smart Waste Segregation System Applying IoT with Notification and Monitoring (Carolino, E. B., et al.,2024) and WasteWise: A Waste Segregation System with Wi-Fi and Charging Reward (De Castro, 2026). Which solves numerous problems such as proper waste awareness and providing Wi-Fi as an incentive. However, MantraCore provides value for both user engagement and waste sorting, giving opportunity for a cycle of user interaction and awareness while providing incentives – promoting healthy operation for both environment and user.

Unlike traditional waste bins, this system provides features that can be beneficial for both users and the environment. Rather than waste bin for multiple different types of waste which users tend to mix different types of waste – this system can segregate materials from paper, plastic and (aluminum cans); which can be used for solving maintenance cost when sold and preventing prolonged biodegrading. The system has a QR-based incentive system for verification of users as security while connecting to their designated accounts to record their activities. In addition, the user can use their accumulated incentives to have discounts on their amenity usage. This system encourages users to properly segregate waste and to be active in the community.

Chapter 4. Startup Opportunity Analysis

4.1 Startup Readiness Assessment

This section evaluates the startup's readiness before proceeding to the Business Development phase. It assesses the startup's problem-solution fit, innovation, market potential, technical feasibility, and team capability to determine its preparedness for implementation and future growth.

Rating Scale

Table 3: Rating Scale

Rating	Interpretation
5	Excellent / Strongly Agree
4	Good / Agree
3	Fair / Neutral
2	Needs Improvement
1	Poor / Strongly Disagree

Assessment Criteria

Table 4: Assessment Criteria

Assessment Criteria	Description
Problem–Solution Fit	The proposed startup effectively addresses a real and significant problem experienced by the target users.
Market Demand	There is evidence that potential customers are interested in the proposed solution.
Innovation	The startup offers unique features or improvements compared with existing solutions.
Technical Feasibility	The proposed technology, tools, and resources are available and can be implemented by the team.
Team Capability	The team possesses the necessary knowledge, skills, and commitment to complete the project successfully.

Individual Assessment Forms:

Evaluator: Bautista, Bianca Nicole M.

Table 5: Individual Assessment Form (Part 1)

Assessment Criteria	Rating (1–5)	Remarks / Justification
Problem–Solution Fit	5	VHEcoPoint directly addresses the critical issue of low resident engagement in proper waste segregation by introducing an automated, rewarding system. This tackles the behavioral root of the problem, which is a key improvement over traditional, passive collection methods.
Market Demand	5	The increasing environmental consciousness and stricter local regulations create a substantial market need. Our interviews proved that both residents and community administrators see clear value in a system that simplifies recycling and offers tangible incentives, positioning us well in a growing sustainability market.

Assessment Criteria	Rating (1–5)	Remarks / Justification
Innovation	5	This project is highly innovative in its integrated approach. By combining machine learning waste detection, a reliable weighing system, a QR-based reward mechanism, and a centralized web platform for data analytics, we are offering a novel, end-to-end solution that is far more engaging and effective than existing alternatives in the local market.
Technical Feasibility	4	While hardware components are not always readily available and can be costly, we have successfully navigated these challenges in our prototyping phase. Our software development is built on a solid foundation of web development skills, making it a viable and achievable goal for our team.
Team Capability	4	As a team, we have shown a strong collective ability to learn and integrate complex technologies for the prototype. Our skills in full-stack development, and systematic testing complement each other well. While we are aware of the challenges like studying thoroughly about the components more. But overall, proven adaptability and solid project management show that we can overcome them.

Evaluator: Catolico, Ayesha Ysabelle R.

Table 6: Individual Assessment Form (Part 2)

Assessment Criteria	Rating (1–5)	Remarks / Justification
Problem–Solution Fit	5	VHEcoPoint directly addresses improper waste segregation by providing an automated recycling station with a QR-based rewards system that encourages residents to recycle responsibly.
Market Demand	5	There is an increasing demand for sustainable waste management solutions in residential communities. The system meets the needs of environmentally conscious homeowners and subdivision administrators.

Assessment Criteria	Rating (1–5)	Remarks / Justification
Innovation	5	The startup combines IoT hardware, AI-based waste detection, QR technology, and a digital incentive system into a single solution, making it more engaging and efficient than traditional recycling methods.
Technical Feasibility	4	The hardware components come at a cost and are not always readily available, but the team can still source and implement them with proper planning. The software, built on the VictorianPass platform, is more readily achievable given the team's existing web development skills.
Team Capability	3	The founding team brings complementary skills in full stack development, IoT engineering, and QA testing, backed by structured project management. As an early-stage startup, however, the team is still building out technical depth and resources compared to established competitors.

Evaluator: Que, Crizzle Anne D.

Table 7: Individual Assessment Form (Part 3)

Assessment Criteria	Rating (1–5)	Remarks / Justification
Problem–Solution Fit	5	The solution directly tackles the core issues of improper disposal and low engagement identified in the Empathize phase. It provides a complete, automated system that addresses both the logistical and behavioral problems.
Market Demand	4	There is a growing demand for smart, sustainable solutions in residential areas. The system's value proposition for convenience, rewards, and data tracking is highly appealing to both residents and administrators as shown in the interview validation.
Innovation	5	The integration of an automated segregation station with a QR-based incentive system and a comprehensive web platform for tracking and analytics is a highly novel and integrated solution not commonly found in the local market.

Assessment Criteria	Rating (1–5)	Remarks / Justification
Technical Feasibility	4	The project uses well-documented and widely available components (ESP32-CAM, GM65, HX711). The prototype proves the concept is workable, and the team has the technical knowledge to proceed with implementation.
Team Capability	3	The team demonstrated the ability to learn and apply new technologies such as Linux OS, YOLOv11, and hardware integration. They successfully delivered a working prototype. However, their expertise is largely academic and may not be sufficient for full-scale deployment, which requires advanced troubleshooting, system security, and long-term maintenance capabilities.

Evaluator: Tabinga, Vincent Yuri Z.

Table 8: Individual Assessment Form (Part 4)

Assessment Criteria	Rating (1–5)	Remarks / Justification
Problem–Solution Fit	5	The VHEcoPoint aims to encourage users to participate in proper waste segregation, with a seamless system to segregate materials from PET bottles, paper and Aluminum cans which can affect and prolonging biodegrading cycle. According to (ScienceInsights,2026),, plastic deration is a process where plastic breakdown overtime – this process speed can depend on the environment conditions such as; sunlight intensity, without exposure will remain intact longer. Plastics can persist up to 450 to 1000 years to fully break down. This system helps the prevention of these kinds of non-biodegradable to mix with biodegradable that can prolong the process of breakdown of matter.
Market Demand	4	The demand on systems that can be used for sustainable and clean communities can interest varied stakeholders, ranging from homeowners to local city councils. This system can deliver awareness to improper waste, and sustainable waste management.
Innovation	5	The system offers weight-based incentives system that the users can interact by inserting Pet bottles, Paper and Aluminum cans while accumulating points for amenity usage

Assessment Criteria	Rating (1–5)	Remarks / Justification
Technical Feasibility	3	The tools and hardware are not easily obtainable due to expenses; however, it is available and can be implemented by the team with trial and error.
Team Capability	2	There are hardware, tools and programming languages that the team needed to first learn before integrating such as Linux OS and network configuration in a Raspberry Pi 4b. Leaning Machine Learning (ML) using YOLOv11 to be streamed on an ESP32-CAM and planning which hardware can be used to build the project.

Group Summary

Table 9: Group Summary

Assessment Criteria	Bautista	Catolico	Que	Tabinga	Average
Problem–Solution Fit	5	5	5	5	5.00
Market Demand	5	5	4	4	4.50
Innovation	5	5	5	5	5.00
Technical Feasibility	4	4	4	3	3.75
Team Capability	4	3	3	2	3.00
Overall Startup Readiness					4.25

Rating Interpretation

Table 10: Rating Interpretation

Average Score	Interpretation
4.50 – 5.00	Highly Ready
3.50 – 4.49	Ready with Minor Improvements
2.50 – 3.49	Needs Further Development

Average Score	Interpretation
Below 2.50	Not Yet Ready

Analysis

Among the five criteria, Problem–Solution Fit and Innovation both received the highest average rating of 5.00, as all four evaluators agreed that VHEcoPoint addresses a genuine, pressing problem while offering something distinct from existing alternatives and tackling the behavioral root cause of poor waste segregation through an automated station paired with a QR-based rewards mechanism, and integrating IoT hardware, AI-based waste detection, and a centralized web platform into one cohesive system that no local competitor currently offers. Market Demand followed closely at 4.50, with evaluators pointing to validated interest from both residents and subdivision administrators during interviews, confirming that the appeal of a smarter, more convenient segregation system is grounded in real feedback rather than assumption. On the other end, Team Capability received the lowest average rating at 3.00 (ranging from 2 to 4 across evaluators), reflecting concerns that while Team Mantra has shown strong ability to learn and adapt and picking up Linux OS, YOLOv11, Raspberry Pi configuration, and network setup and much of this expertise was built in real time during the project rather than something the team already possessed, with Tabinga specifically flagging that ML integration on the ESP32-CAM had to be learned from scratch. Technical Feasibility scored similarly modest at 3.75, mainly due to the cost and limited availability of hardware components, though the team's successful prototyping suggests this is manageable rather than fundamental. These ratings are supported by three main sources of evidence: interview validation with residents and subdivision administrators, a working prototype from the Empathize and Design phases, and a comparative analysis showing no direct competitor offers the same end-to-end integration of hardware, incentives, and analytics.

With an overall average of 4.25, the startup falls under "Ready with Minor Improvements," meaning Team Mantra can proceed to the Business Development phase, but only after addressing the gaps in Team Capability and Technical Feasibility and the core concept, market need, and innovation are already solid and well-validated. Before the Midterm Startup Validation Defense, the team should prioritize closing skills gaps around IoT security, system troubleshooting, and long-term maintenance planning, document a clearer hardware sourcing and cost-contingency plan to address feasibility concerns, and compile the interview and validation data into a more structured evidence base for a stronger, more confident presentation.

4.2 Market and Competitor Analysis

Target Market

The target market of VHEcoPoint is primarily residential communities, particularly subdivisions and Homeowners' Associations (HOAs) that experience challenges in waste segregation, recycling participation, and waste monitoring. The initial target market is Victorian Heights Subdivision, where the system aims to provide a more organized and technology-assisted way of managing recyclable materials.

The system focuses on residents and homeowners as direct users, while HOAs and subdivision administrators benefit from monitoring recycling activities and resident participation through VictorianPass. VHEcoPoint encourages residents to properly segregate recyclable materials through QR-based identification and incentives, while providing administrators with useful records for managing the program.

In the future, VHEcoPoint can be introduced to other subdivisions, gated communities, and HOAs with similar waste management needs. This gives MantraCore Solutions an opportunity to expand VHEcoPoint as a practical and scalable smart waste management solution for residential communities.

Primary Customers:

The primary customers are the individuals who are the primary main target of MantraCore's VHEcoPoint that can interact and benefit from the system features such as; incentives. They are the central focus of MantraCore's knowledge and skills to solve their ongoing problems, which is the group's needs, preferences and behavior is a priority as they build a long-term success of the VHEcoPoint. The primary customers of VHEcoPoint are the homeowners or residents of Victorian Heights Subdivision. They are the direct users who will interact with the station by scanning their QR code and depositing recyclable materials. The system is designed to make recycling more convenient while encouraging residents to develop proper waste segregation habits.

Residents fro Victorian Heights Subdivision can benefit from the system by earning reward points based on their recycling activities. These points may be used for available community incentives, giving residents an additional reason to participate regularly. Through this approach, VHEcoPoint connects responsible waste disposal with direct benefits for residents.

Secondary Customers:

The secondary customers are the Homeowners' Association (HOA), Maintenance staff and guard. They are the individuals who influence the decision-making process, as they help shape potential buyers and users of the system even if they do not directly interact with it. However, enables and can be beneficiaries of the product presented. Understanding their behavior, influence and drive to ensure that the system can be adopted in a real-life setting is crucial in deployment and can be used as opportunity for possible revenue. ignoring their perception can result in lost or weaker system satisfaction deployment. Recorded information can also help developers evaluate the effectiveness of the system prior to deployment and make better decisions based on actual participation and recycling records. Ensuring VHEcoPoint can effectively deployed and helpful not only for residents but also for the overall management of the community's sustainability efforts.

Competitor Analysis

The market for smart waste solutions is emerging but has both indirect and direct competitors. The following analysis evaluates competitors based on the features and value proposition of the MantraCore Solutions project.

Table 11: Competitor Analysis

Competitor	**Strengths**	**Weaknesses**
Competitor 1: Barangay/Informal Waste Collectors	Existing and familiar service. Low-cost or free for residents	No incentive program for residents. No data tracking or reporting for administrators.
Competitor 2: Traditional Manual Segregation Bins	Low initial cost. Simple to deploy. Familiar concept to residents.	Heavily relies on residents to sort waste correctly, which is the core problem. No system to track individual participation. No reward or incentive mechanism. No data collection for performance analysis.
Competitor 3: Other Smart Waste Startups (e.g., Basic IoT Bins, Reward Apps)	Integrate technology for waste management. May offer some form of reward system or data tracking. Modern and innovative image	Lack of local support and maintenance – Foreign systems often have no local technical support, making repairs and troubleshooting difficult and costly. May lack an integrated hardware-software approach. May be expensive to implement or have complex user interfaces. May not be specifically tailored to the needs of Philippine residential subdivisions.

Market Opportunity

There is a clear and significant gap in the market for a community-integrated smart waste segregation solution that simultaneously automates material sorting, verifies resident identity, rewards proper disposal behavior, and provides administrators with real-time monitoring all within a single platform. Existing solutions address only one or two of these needs in isolation. Traditional junk shops offer cash but no monitoring. Smart bins offer monitoring but no resident engagement. EcoWaste programs offer awareness but no automation. MantraCore Solutions addresses this gap by integrating VHEcoPoint with the VictorianPass platform, enabling residents to earn points redeemable for free amenity bookings while giving subdivision administrators full visibility into recycling activity creating a solution that is both community-driven and data-informed.4

4.3 Technology Trends

The market gap identified in the previous section is reinforced by several emerging technology trends. Several technology trends currently support the viability of MantraCore Solutions:

1. Rise of IoT in Smart City and Smart Community Solutions – The growing adoption of Internet of Things (IoT) devices in residential and urban environments enables real-time monitoring, automation, and data collection at the community level, moving beyond traditional standalone appliances.
2. QR Code-Based Identification and Contactless Verification – QR technology has become a widely accepted, low-cost method for identity verification and transaction tracking, especially following its mainstream adoption during the pandemic for payments, check-ins, and access control.
3. Growing Emphasis on Environmental, Social, and Governance (ESG) Compliance – Homeowners associations, real estate developers, and local government units face increasing pressure to demonstrate sustainable practices, creating demand for technology that can measurably support waste diversion and environmental reporting.
4. Affordable Microcontroller and Sensor Technology – The declining cost and increasing availability of components such as the Raspberry Pi 4b, ESP32-CAM, HX117 loadcells module, and QR scanners have made IoT-based automation financially viable for small startups and community-scale deployments, rather than being limited to large-scale industrial applications.

Discussion

These trends directly support MantraCore Solutions' value proposition and reinforce the market opportunity identified in the competitor analysis. The rise of IoT adoption at the community level establishes a receptive environment for a system like VHEcoPoint, since residents and administrators are already growing accustomed to connected devices that automate everyday tasks and generate usable data, rather than treating smart technology as a novelty.

The convergence of affordable IoT hardware and QR-based identification allows the company to build an automated, verifiable waste segregation system without the high capital cost historically associated with industrial-grade waste automation. Components such as the ESP32-CAM, GM65 QR scanner, and HX711 load cell make it possible to deliver real-time material detection, resident verification, and weight-based point computation at a cost structure that a subdivision-scale deployment can realistically afford.

At the same time, the mainstream familiarity with QR codes lowers the learning curve for residents. Because QR scanning is already embedded in everyday transactions such as payments and check-ins, residents are not being asked to learn an unfamiliar technology; they are simply applying a familiar interaction to a new context, which reduces resistance to adoption and shortens the time needed for behavior change.

The growing acceptance of gamified incentive systems further strengthens this adoption pathway. Residents who are already accustomed to points-and-rewards mechanics in other digital platforms are more receptive to a similar model for waste segregation, which lowers the barrier to sustained participation rather than requiring the company to introduce an entirely new behavioral concept.

Finally, the increasing ESG expectations placed on subdivisions and HOAs strengthen the business case for administrators to adopt a system like VHEcoPoint, since it gives them a concrete, data-backed way to demonstrate environmental responsibility to residents, developers, and local government. As compliance and reporting pressures continue to grow, administrators are more likely to view an automated tracking system not as an optional convenience, but as a practical tool for meeting emerging governance standards.

4.4 Unique Selling Proposition (USP)

USP Statement

MantraCore Solutions provides an automated, QR-based smart waste segregation and incentive system to homeowners' associations and residential communities by offering real-time material sorting, verified resident tracking, and redeemable rewards for amenity usage, making it more integrated and community-tailored than existing manual bins, informal collectors, or foreign smart waste systems.

Key Competitive Advantages

1. Fully integrated hardware-software ecosystem – Unlike manual segregation bins or basic smart bins, VHEcoPoint combines automated material classification, resident verification, and a web-based platform (VictorianPass) into a single seamless system, rather than requiring residents or administrators to rely on separate, disconnected tools.
2. Locally tailored design and support – As a Philippine-developed solution built specifically around the operational realities of subdivisions and HOAs, MantraCore Solutions offers accessible local support and a system designed around local waste patterns, unlike foreign smart waste products that often lack local maintenance and troubleshooting.
3. Tangible, redeemable incentives tied to community life – Points earned through recycling can be directly redeemed for amenity usage (e.g., clubhouse, multi-purpose buildings, courts) within the residents own subdivision, creating a direct and meaningful incentive loop that generic reward-point apps or basic smart bins do not offer.

Chapter 5. Customer Validation

5.1 Customer Persona

Persona 1: Resident

GAILE

PROFILE

Full Name : Gaile Sarmiento
Age : 46
Occupation : IT Programmer at GMA
Location : Ciudad Real Subdivision, San Jose Del Monte Bulacan
Income Level : Upper Middle Class

NEEDS

- A more efficient collection schedule to prevent waste accumulation at home
- Separate collection days for biodegradable and non-biodegradable waste
- A system that classifies recyclables into specific categories (plastic containers, glass bottles, paper) for easier recycling

PAIN POINTS

- Garbage collection is only twice a week, causing kitchen waste to pile up quickly
- Accumulated biodegradable waste attracts rats and pests, posing health risks to her family
- Both biodegradable and non-biodegradable are collected on the same day, making her segregation efforts feel pointless
- Limited space at home to store waste for several days, especially with children around

BUYING BEHAVIOR

- Environmentally conscious and family-oriented; prioritizes cleanliness and safety
- Wants a system that is practical, convenient, and reduces health hazards at home
- Supportive of initiatives that make recycling easier and more rewarding
- Motivated by solutions that protect her children from pests and disease

Figure 4: Persona 1

Persona 2: Admin Assistant

GIGI M. CARANDANG

PROFILE

Full Name : Gigi M. Carandang
Age : 51
Occupation : Admin Assistant
Location : New Haven
Income Level : Middle Class

NEEDS

- A more efficient waste management system that reduces reliance on manual segregation by garbage men
- Clearer and more effective communication strategies to ensure resident compliance
- A system that simplifies waste handling and minimizes staff workload

PAIN POINTS

- Despite multiple announcements, some residents still do not follow segregation rules
- Garbage men are overburdened with having to re-segregate waste that was not properly sorted by residents
- Repeated violations without consequences make enforcement difficult

BUYING BEHAVIOR

- Practical and solution-oriented; prefers systems that directly address operational inefficiencies
- Supportive of initiatives that ease the workload of staff and improve community cleanliness
- Values long-term benefits over short-term convenience, especially for the welfare of workers

Persona 3: Security Guard

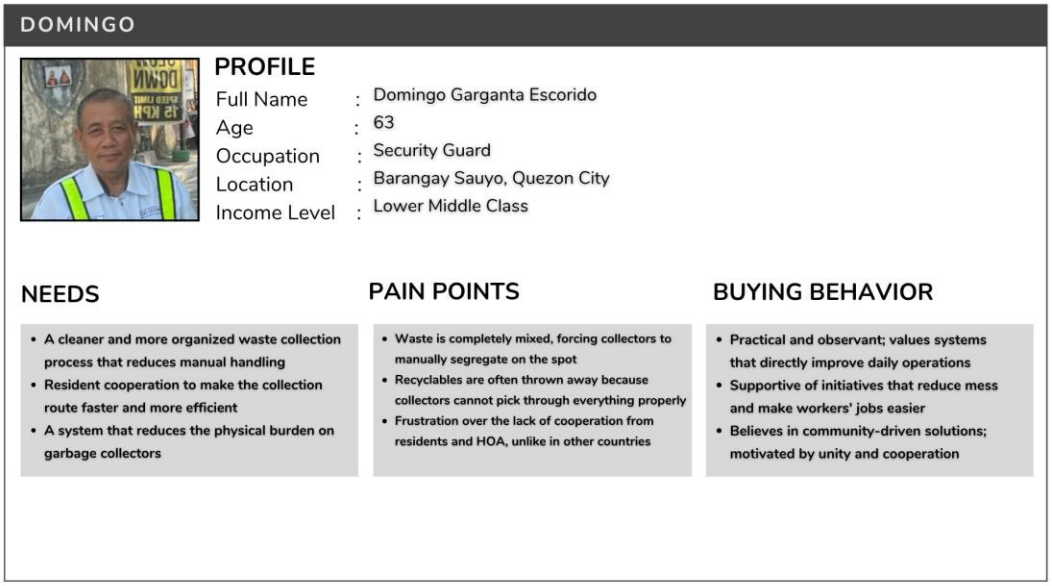

Figure 6. Persona 3

Persona 4: Resident

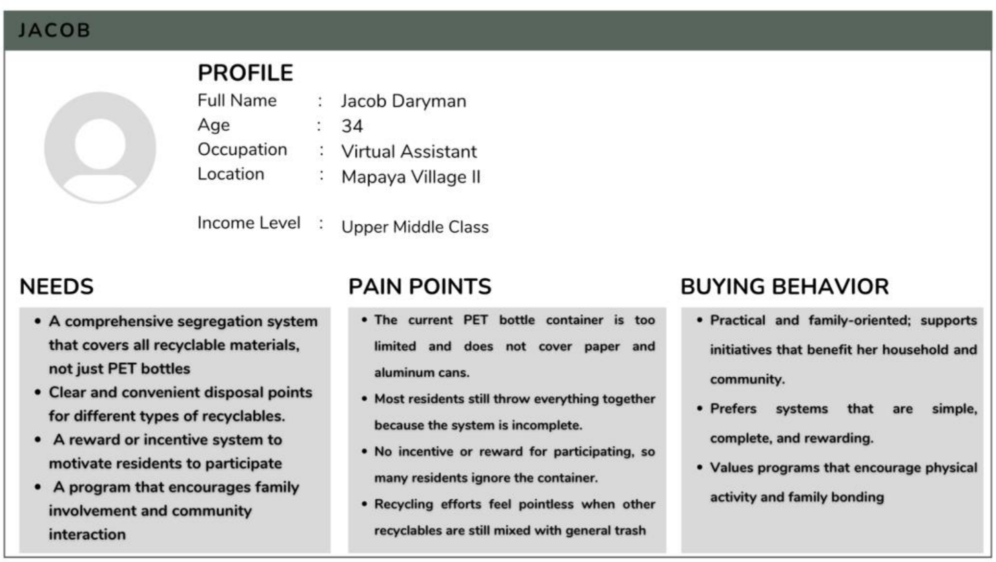

Figure 7: Persona 4

Persona 5: Resident

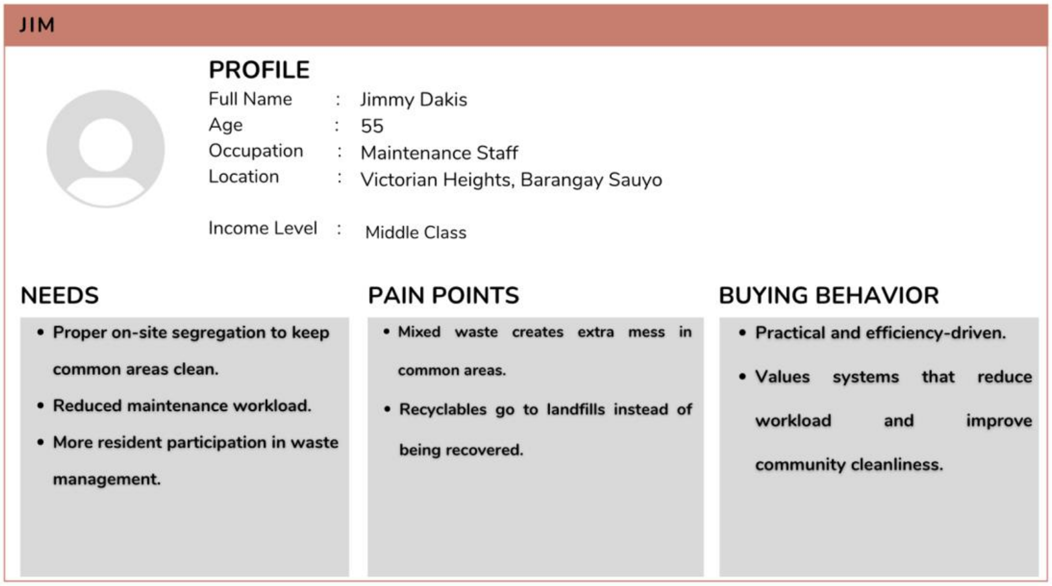

Figure 8: Persona 5

5.2 Interview Results

The following questions were designed to be neutral and unbiased, avoiding leading language that might influence respondent answers:

1. Can you describe how waste, particularly recyclable materials like bottles, paper, and aluminum cans, is currently managed in your community? And What methods or practices on your community currently use to handle recyclable waste?
2. What difficulties or frustrations do you face with the current waste management practices in your subdivision?
3. Would you recommend this product VHEcoPoint: A Smart Waste Segregation Station with QR-Based Incentive System for Victorian Heights to other HOA or communities? Why?

Interview Summary Table

Table 12: Interview Summary Table

Respondent	**Role**	**Key Feedback**	**Interested?**
Respondent 1	Admin Assistant (New Haven Village)	Residents currently segregate biodegradable/non-biodegradable into public trashcans, but garbage men still handle the final sorting. The main challenge is resident non-compliance despite repeated announcements. Recommends VHEcoPoint because it would make the garbage men's jobs much easier and simplify the process for Filipinos.	Yes
Respondent 2	Security Guard (Victorian Heights Subd.)	Zero sorting from residents makes collectors' work messy and time-consuming, causing recyclables to get thrown away. Highly recommends it because pre-sorting would make staff work cleaner/faster and would strengthen the subdivision's sense of community and unity.	Yes
Respondent 3	Resident (Cuidad Real Subdivision)	Residents currently segregate biodegradable and non-biodegradable into different plastics. Main issues: collection is only twice a week (accumulating kitchen waste and attracting pests) and both types are collected on the same day, undermining the segregation effort. Recommends VHEcoPoint to collect biodegradables separately and to classify recyclables into specific bags for plastics, glass bottles, and paper to make recycling easier.	Yes
Respondent 4	Maintenance Staff (Victorian Heights Subd.)	Mixed waste sends recyclables to landfills and creates messy, dirty common areas. Endorses the product because proper on-site segregation improves community cleanliness, gets residents active, and significantly reduces the maintenance staff's workload.	Yes

Respondent	Role	Key Feedback	Interested?
Respondent 5	Resident (Mapayapa Village II)	Current recycling is limited to PET bottles with no incentives, resulting in low participation. Recommends VHEcoPoint because it covers all recyclables, uses a reward system to actually drive participation, and encourages communal outdoor walking.	Yes

5.3 Validation Findings

The customer interviews provided overwhelmingly positive validation for the VHEcoPoint concept, with all five respondents expressing a clear "Yes" to recommending the product to other communities, demonstrating a strong market demand. The most compelling validation came from the residents' recognition that a QR-based reward system would serve as a powerful daily motivator to encourage proper segregation, while multiple respondents also highlighted the unexpected social and health benefits of the station, noting it would encourage residents to walk more and strengthen neighborhood ties through increased interaction. The staff respondents validated the product for its operational value, emphasizing that pre-sorting waste would significantly ease the physical burden on garbage collectors and maintenance workers, reduce mess in common areas, and prevent valuable recyclables from ending up in landfills. However, the interviews also validated the startup by exposing critical gaps that VHEcoPoint must address, such as the lack of specific bins for plastics, glass, and paper, the infrequency of garbage collection schedules that attract rats, the residents' resistance to following existing rules, and the current limitation of recycling efforts to only PET bottles. Overall, the customers strongly validated the idea not merely as an optional amenity, but as a necessary solution to a real problem they face daily, making it clear that the product fulfills a genuine practical need while also creating a cleaner, healthier, and more connected community.

5.4 Recommended Improvements

Based on the specific frustrations and feedback from the respondents, the following improvements are recommended for the VHEcoPoint system:

1. **Implement Multi-Category Sorting Compartments:** Instead of a single "Recyclable" bin, the station should feature distinct, clearly labeled compartments specifically for **PET Bottles, Aluminum Cans, and Paper**. This directly addresses the feedback from Respondent 3 and ensures recycling centers receive pre-sorted materials, reducing the need for further sorting.

2. **Establish a Separate Biodegradable Drop-Off:** As suggested by Respondent 3, the station could include a separate, secured bin specifically for kitchen/biodegradable waste. This would allow residents to safely dispose of food scraps outside of the limited 2x-a-week collection schedule, effectively reducing the rat and odor issue.

3. **Design User-Friendly Bags/Containers for Removal:** To achieve the goal of making garbage men's jobs easier (Respondent 1), the station's internal bins should use easy-to-pull, durable, color-coded trash bags that collectors can simply tie and swap out without needing to manually sort inside the station again.
4. **Launch with an Educational Signage Campaign:** Since Respondent 1 noted resident non-compliance despite multiple announcements, the launch of VHEcoPoint should be accompanied by highly visible, visual infographic signage on the station itself. These should precisely show exactly *which* items go into which compartment, coupled with a "Sorting Guide" that residents can scan via the QR code to earn a small initial reward.
5. **Coordinate with the HOA on Collection Logistics:** If possible, the project team should recommend to the HOA that the newly segregated, recyclable waste from VHEcoPoint be collected on a different schedule (or by a different recycler) than the standard biodegradable truck, ensuring that the residents' segregation efforts at the station do not get mixed back together on the same pickup day.

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